

**VISVESVARAYA TECHNOLOGICAL UNIVERSITY  
JNANA SANGAMA, BELAGAVI – 590018**



*A Synopsis on*

**“APPLICATION OF A BEAMFORMING TECHNIQUES ON  
FOR  
FIELD SPEECH RECOGNITION FOR REGIONAL  
LANGUAGE”**

*Submitted for the Major Project Work-Phase 1 of the Academic Year 2026-27*

**Submitted by**

|                                         |                     |
|-----------------------------------------|---------------------|
| <b>Nandan Kuchabal</b>                  | <b>(1NT23EC090)</b> |
| <b>Sagar A S</b>                        | <b>(1NT23EC134)</b> |
| <b>Relangi Mahesh Satya Venkat Gowd</b> | <b>(1NT23EC127)</b> |

*Under the Guidance of*

**Prof Amruth Ashok Gadag**  
Assistant Professor

**Department of Electronics and Communication Engineering**



**An Autonomous Institution Affiliated to Visvesvaraya Technological University, Yelahanka, Bangalore-  
560064, Karnataka, INDIA**

**AY 2026-2027**



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

## *Certificate*

This is to certify that the **Phase-1** project work entitled “**APPLICATION OF A BEAMFORMING TECHNIQUES ON FOR FIELD SPEECH RECOGNITION FOR REGIONAL LANGUAGE**” has been carried out by *Nandan Kuchabal (INT23EC090)*, *Sagar A S (INT23EC134)*, and *Relangi Mahesh Satya Venkat Gowd (INT23EC127)*, bonafide students of *Nitte Meenakshi Institute of Technology*, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in the **Department of Electronics and Communication Engineering** under **Visvesvaraya Technological University**, Belagavi, during the academic year **2026–2027**.

The project report has been examined and approved as it meets the academic requirements specified under the autonomous scheme of **Nitte Meenakshi Institute of Technology** for the said degree.

Signature of the Guide

Signature of the HoD

Signature of the Principal

**Prof. Amruth Ashok Gadag**

Assistant Professor  
Dept. of ECE, NMIT

**Dr. Parameshachari B D**

HoD  
Dept. of ECE, NMIT

**Dr. H C Nagaraj**

Principal  
NMIT

## *External Viva-Voce*

*Name of Examiners*

*Signature with Date*

1. ....

.....

2. ....

.....

## Acknowledgement

The successful execution of our project has been a significant milestone, and we take this opportunity to express our heartfelt gratitude to all those who have supported and guided us throughout this journey. Whatever we have achieved is the result of their encouragement and help, and we remain deeply thankful to each one of them.

We also convey our deep appreciation to **Dr. Parameshachari B D, Professor and HoD**, *Department of Electronics and Communication Engineering*, for his constant encouragement, valuable guidance, and for fostering a vibrant learning environment. We extend our heartfelt thanks to our project guide, **Prof. Amruth Ashok Gadag**, Assistant Professor, *Department of Electronics and Communication Engineering* for his/her unwavering support, timely feedback, and insightful mentorship throughout the project. We are also indebted to our parents for their unconditional love, support, and encouragement throughout our academic journey at *Nitte Meenakshi Institute of Technology*, Bengaluru. Finally, we would like to express our appreciation to all those—**named and unnamed**—who contributed in any way to our learning and success.

|                                  |              |
|----------------------------------|--------------|
| Nandan Kuchabal                  | (1NT23EC090) |
| Sagar A S                        | (1NT23EC134) |
| Relangi Mahesh Satya Venkat Gowd | (1NT23EC127) |

Place: NMIT, Bengaluru

## Academic Year: 2026–2027

### Declaration by Students

We, the undersigned students **Nandan Kuchabal (1NT23EC090)**, **Sagar A S(1NT23EC134)**, **Relangi Mahesh Satya Venkat Gowd (1NT23EC127)** hereby declare that we are fully aware of the **project conduction procedures** and **evaluation rubrics** as prescribed by the department. We understand that the **project work will be evaluated through continuous assessment** across the defined phases.

| <i>Name of the Student</i>       | <i>Signature</i> |
|----------------------------------|------------------|
| Nandan Kuchabal                  |                  |
| Sagar A S                        |                  |
| Relangi Mahesh Satya Venkat Gowd |                  |

### Confirmation by Guide

I, **Amruth Ashok Gadag** hereby declare that I shall guide the above-mentioned students in the project entitled “**APPLICATION OF A BEAMFORMING TECHNIQUES ON FOR FIELD SPEECH RECOGNITION FOR REGIONAL LANGUAGE**”and will ensure that the project work is carried out under my supervision and submitted as per the academic requirements for the academic year **2026–2027**.

*Signature of the Guide*

## TABLE OF CONTENTS

| Sr. No. | Contents                     | Page No. |
|---------|------------------------------|----------|
| 1       | Introduction                 | 1        |
| 2       | Objectives of the Project    | 2        |
| 3       | Literature Survey and Review | 3        |
| 4       | Gap Analysis and Motivation  | 4        |
| 5       | Functional Requirements      | 5        |
| 6       | Proof of Concept             | 6        |
| 7       | Methodology                  | 7        |
| 8       | System Architecture          | 8        |
| 9       | Algorithms Used              | 9        |
| 10      | Dataset Description          | 10       |
| 11      | Experimental Setup           | 11       |
| 12      | Results and Analysis         | 12       |
| 13      | Future Scope                 | 13       |
| 14      | Conclusion                   | 14       |
| 15      | References                   | 15       |

# 1. Introduction

---

Speech recognition technology has become an essential part of modern human–computer interaction systems. Applications such as virtual assistants, smart devices, automated customer support systems, online meeting platforms, and voice-controlled interfaces depend on the ability of machines to accurately understand spoken language. In recent years, Automatic Speech Recognition (ASR) systems have achieved significant progress because of advancements in artificial intelligence, deep learning, and signal processing techniques. However, the performance of these systems is still highly dependent on the quality of the captured speech signal.

In practical environments, speech is often recorded from a distance rather than through close-talk microphones. Such speech acquisition conditions are commonly referred to as far-field speech environments. In these situations, the recorded signal becomes degraded due to background noise, room reverberation, echo, competing speakers, and other environmental disturbances. These factors reduce speech clarity and negatively affect the accuracy of speech recognition systems. Therefore, improving speech quality before recognition becomes an important requirement in far-field ASR applications.

To overcome these challenges, beamforming techniques are widely used in multi-microphone speech processing systems. Beamforming is a spatial filtering method that uses signals captured from multiple microphones to focus on sound arriving from a desired direction while suppressing unwanted noise and interference from other directions. By improving the signal-to-noise ratio (SNR) of the captured speech, beamforming helps enhance speech intelligibility and improves the reliability of recognition systems in noisy environments.

This project focuses on the application of beamforming techniques for improving far-field speech recognition in regional languages, specifically Kannada. Compared to globally dominant languages, regional languages often have limited speech datasets, fewer optimized recognition models, and reduced research attention in real-world noisy conditions. Developing robust far-field speech recognition systems for regional languages is therefore important for improving accessibility, inclusiveness, and the usability of voice-enabled technologies for a wider population.

The proposed work aims to design a prototype framework that combines microphone array signal processing, beamforming-based speech enhancement, and speech recognition techniques to improve the recognition performance of distant regional-language speech. The study mainly investigates how preprocessing methods such as Delay-and-Sum beamforming and speech enhancement can improve the quality of noisy speech before it is processed by an ASR model. The effectiveness of the system will be analyzed by comparing recognition performance before and after beamforming enhancement.

The outcome of this work can contribute towards the development of practical voice-based systems capable of operating effectively in real-world environments such as classrooms, public service centers, smart homes, and assistive communication applications where distant speech acquisition is common.

## 2. Objectives of the Project

---

1. To study the challenges associated with far-field speech recognition systems in noisy and reverberant environments.
2. To understand the working principles of microphone arrays and beamforming techniques used for speech enhancement applications.
3. To implement Delay-and-Sum (DS) beamforming for improving the quality of speech signals captured from distant speakers.
4. To reduce the effect of background noise, reverberation, and interference present in far-field speech recordings.
5. To preprocess and enhance regional-language speech signals before feeding them to the speech recognition model.
6. To apply an Automatic Speech Recognition (ASR) model for converting enhanced Kannada speech into text.
7. To compare the recognition performance of raw speech signals and beamformed speech signals using qualitative and quantitative observations.
8. To evaluate the feasibility of developing a practical far-field speech recognition framework for regional-language applications.

### 3. Literature Survey and Review

---

Research in the field of far-field speech recognition has increased significantly in recent years because speech signals captured from distant speakers are highly affected by background noise, reverberation, and environmental interference. These challenges reduce the accuracy of Automatic Speech Recognition (ASR) systems and make speech enhancement an important preprocessing stage before recognition.

Several researchers have studied the use of microphone array processing and beamforming techniques to improve speech quality in noisy environments. Beamforming uses multiple microphones to capture spatial information about incoming sound signals and enhances speech coming from a desired direction while suppressing unwanted noise from other directions. Because of its effectiveness, beamforming has become one of the most widely used approaches for distant speech enhancement.

Traditional beamforming methods such as Delay-and-Sum (DS) beamforming and Minimum Variance Distortionless Response (MVDR) beamforming are commonly used in far-field speech processing systems. Delay-and-Sum beamforming improves speech quality by aligning and averaging signals from multiple microphones, whereas MVDR beamforming further minimizes noise and interference while preserving the desired speech signal. Many studies have shown that these techniques improve the signal-to-noise ratio (SNR) and increase speech intelligibility in practical environments.

Recent developments in speech technology have also introduced deep learning-based beamforming and neural speech enhancement approaches. Researchers have combined beamforming with deep neural networks and Transformer-based ASR systems to improve recognition performance in highly noisy and reverberant environments. Modern ASR frameworks such as OpenAI Whisper and end-to-end neural architectures have demonstrated better robustness compared to conventional speech recognition methods.

Several benchmark studies including the CHiME challenge series focused on realistic far-field speech recognition tasks. These studies demonstrated that combining beamforming, dereverberation, and neural acoustic modeling significantly improves recognition accuracy in multi-speaker and noisy environments. The results from these studies highlight the importance of front-end speech enhancement techniques for practical ASR systems.

In the Indian context, organizations such as AI4Bharat and projects like IndicVoices have contributed towards the development of multilingual and regional-language speech datasets. These resources support research in Indian language ASR systems, including Kannada speech processing. However, most of the available research mainly focuses on general speech recognition, while limited work has been reported specifically for beamforming-based far-field speech recognition in regional languages.

#### Review of Existing Work

Based on the surveyed literature, the following observations can be made:

1. Far-field speech recognition performance decreases significantly because of background noise, reverberation, and distant speech capture conditions.



2. Beamforming techniques using microphone arrays are effective in improving speech quality and suppressing unwanted interference.
3. Delay-and-Sum and MVDR beamforming methods are among the most commonly used techniques for far-field speech enhancement.
4. Combining speech enhancement techniques with deep learning-based ASR models provides better recognition accuracy in noisy environments.
5. Existing research mainly focuses on English and other high-resource languages, while regional-language far-field ASR systems still require more attention.
6. There is a need for practical prototype-level systems that combine beamforming and ASR for regional-language applications such as Kannada speech recognition.

## 4. Gap Analysis and Motivation

---

### Limitations of Existing Work (Gap Analysis)

Although significant progress has been made in the field of far-field speech recognition, several limitations can still be identified in existing research and practical systems.

1. Most existing systems focus mainly on high-resource languages.  
Many research works and commercial speech recognition systems are primarily developed for languages such as English and other globally dominant languages. Regional languages like Kannada receive comparatively less attention, resulting in limited datasets, fewer optimized models, and reduced research contributions for practical far-field applications.
2. Existing studies often rely on controlled environments.  
Several speech recognition systems are evaluated under laboratory conditions where background noise and reverberation are limited. However, real-world environments such as classrooms, public spaces, and homes contain varying noise levels and acoustic disturbances that reduce recognition performance.
3. Complex hardware requirements increase system cost and implementation difficulty.  
Some beamforming-based systems require advanced microphone arrays and high-end processing hardware, making them difficult to deploy for educational, low-cost, or prototype-level applications.
4. Limited integration between beamforming and regional-language ASR.  
Many existing studies focus either on speech enhancement or speech recognition independently. Fewer works combine beamforming-based speech enhancement with regional-language speech recognition in a unified practical framework.
5. Performance degradation in noisy and reverberant conditions still remains a challenge.  
Even though beamforming improves speech quality, recognition accuracy may still decrease when the environment contains strong reverberation, multiple speakers, or unpredictable background interference.
6. Insufficient prototype-level research for student and academic applications.  
There is a need for simplified and practical implementations that demonstrate the feasibility of beamforming-based far-field ASR systems using accessible software tools and datasets.

### Motivation

The motivation for this project comes from the increasing demand for reliable voice-based communication systems that can function effectively in practical real-world environments. Modern applications such as virtual assistants, smart devices, online meeting systems,

and assistive technologies require speech recognition systems that can accurately process speech even when the speaker is located far from the microphone.

In far-field environments, speech signals are heavily affected by noise, echo, reverberation, and interference from surrounding sounds. These disturbances reduce speech clarity and negatively impact recognition accuracy. Beamforming provides a promising solution by enhancing the desired speech signal using spatial filtering techniques with multiple microphones.

Another important motivation for this work is the limited research available for regional-language far-field speech recognition. While speech technologies for high-resource languages have advanced rapidly, regional languages such as Kannada still require improved datasets, enhancement techniques, and practical ASR frameworks. Developing robust speech recognition systems for regional languages can support better digital accessibility and encourage inclusive technology development.

This project is therefore motivated by the need to design a practical prototype that combines beamforming-based speech enhancement and ASR techniques for improving far-field speech recognition performance in Kannada language applications.

## 5. Functional Requirements

---

The functional requirements define the important operations that the proposed prototype system must perform during Major Project Phase-1. These requirements focus on validating the feasibility of beamforming-based far-field speech recognition for regional-language applications.

| Req ID | Functional Requirement                                                                                                                               | Expected Outcome                                                                  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| FR1    | The system shall acquire far-field speech input in the Kannada language using a simulated multi-microphone array setup or prerecorded audio samples. | Speech signals are successfully captured for further processing and analysis.     |
| FR2    | The system shall preprocess the captured speech signals by applying noise simulation and normalization techniques before enhancement.                | The speech signal becomes suitable for beamforming and ASR processing.            |
| FR3    | The system shall implement Delay-and-Sum beamforming using multiple microphone signals to enhance the desired speech and suppress background noise.  | Enhanced speech with improved clarity and reduced interference is generated.      |
| FR4    | The system shall apply bandpass filtering to remove unwanted low-frequency and high-frequency noise components from the speech signal.               | The processed speech signal contains improved spectral quality and reduced noise. |
| FR5    | The system shall convert the enhanced Kannada speech signal into text using an Automatic Speech Recognition (ASR) model.                             | Recognized Kannada text is generated from the processed speech input.             |
| FR6    | The system shall compare the recognition performance before and after beamforming enhancement.                                                       | The effectiveness of beamforming-based preprocessing is evaluated.                |
| FR7    | The system shall display comparative observations and analysis results for raw speech and enhanced speech signals.                                   | Quantitative and qualitative evaluation results are obtained for the prototype.   |

## 6. Proof of Concept

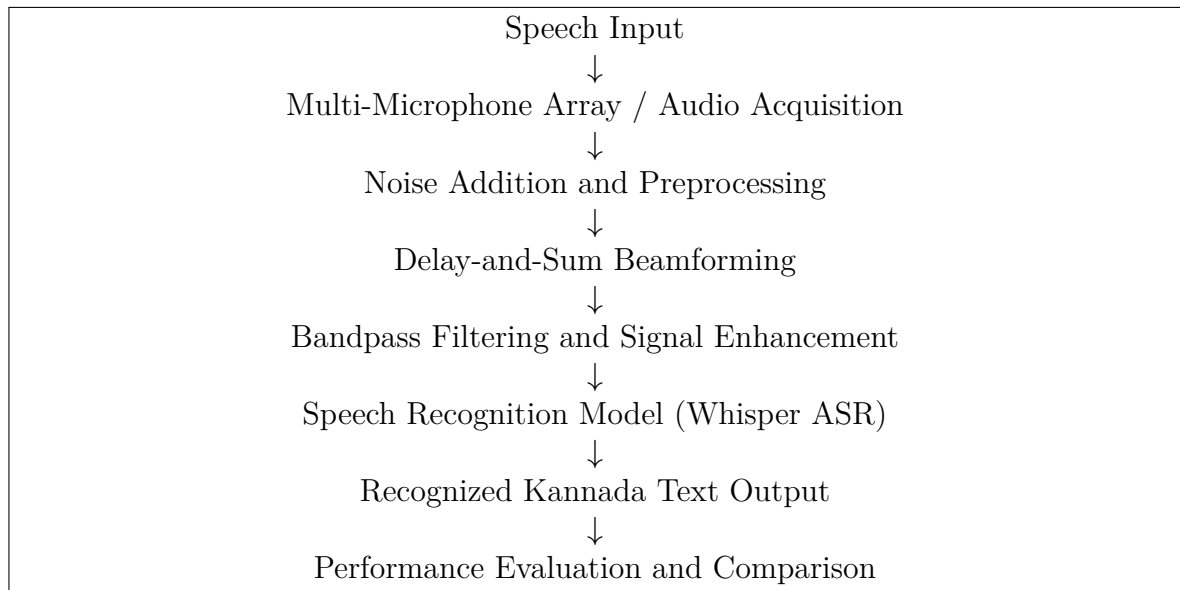
---

The proposed Proof of Concept (PoC) demonstrates the feasibility of improving far-field speech recognition using beamforming techniques for regional-language speech processing. The prototype focuses on enhancing distant speech signals captured in noisy environments before they are processed by an Automatic Speech Recognition (ASR) system.

The proof of concept is designed as a prototype-level implementation that combines multi-microphone speech acquisition, beamforming-based speech enhancement, signal preprocessing, and speech recognition into a single processing pipeline. The system mainly investigates how beamforming can improve the clarity and recognition performance of Kannada speech recorded under far-field conditions.

### 6.1 System Architecture / Block Diagram

The proposed system consists of multiple stages beginning from speech acquisition to final text generation. A simulated microphone array captures the speech signal, which is then processed using beamforming and filtering techniques before being provided to the ASR model.



### 6.2 Workflow / Process Steps

The proof of concept operates through the following sequence of steps:

Step 1: Kannada speech audio is collected using prerecorded samples or microphone input.

Step 2: The clean speech signal is converted into simulated far-field speech by adding background noise and interference.

Step 3: Multiple microphone channels are generated to simulate a microphone

array setup.

- Step 4: Delay-and-Sum beamforming is applied to enhance the desired speech signal and reduce environmental noise.
- Step 5: The beamformed speech signal is further processed using bandpass filtering and normalization techniques.
- Step 6: The enhanced speech signal is provided as input to the Whisper Automatic Speech Recognition model configured for Kannada language processing.
- Step 7: The recognized text output is generated and compared with the output obtained from the noisy speech signal.
- Step 8: Performance observations are analyzed to evaluate the effectiveness of beamforming for far-field speech recognition.

## 6.3 Hardware and Software Requirements

### Hardware

- Computer or Laptop System
- Simulated Multi-Microphone Array Setup
- Audio Input Device / Microphone
- GPU-enabled environment for ASR inference (optional)

### Software

- Python Programming Language
- Google Colab / Jupyter Notebook
- OpenAI Whisper ASR Model
- NumPy and SciPy Libraries
- Librosa for Audio Processing
- Matplotlib for Visualization
- FFmpeg and SoundFile Libraries

## 6.4 Techniques Implemented

The following techniques are implemented in the prototype system:

1. Speech Signal Preprocessing
2. Noise Simulation for Far-Field Conditions
3. Delay-and-Sum Beamforming

4. Bandpass Filtering
5. Signal Normalization
6. Automatic Speech Recognition using Whisper
7. Comparative Performance Evaluation

## 6.5 Expected Outcome

The proposed proof of concept is expected to demonstrate that beamforming-based speech enhancement improves the quality and intelligibility of far-field Kannada speech signals. The enhanced speech is expected to provide more stable and accurate speech recognition results compared to raw noisy speech signals. The prototype also aims to validate the feasibility of developing practical regional-language far-field speech recognition systems using beamforming techniques.

## 7. Methodology

---

The proposed methodology focuses on developing a prototype framework for improving far-field speech recognition using beamforming techniques for Kannada language speech. The system combines speech signal preprocessing, microphone array simulation, beamforming-based enhancement, and Automatic Speech Recognition (ASR) into a single processing pipeline.

The overall methodology is divided into multiple stages beginning from speech data acquisition to final performance evaluation. Each stage contributes towards improving the quality of distant speech signals before they are processed by the recognition model.

### 7.1 Stage 1: Speech Data Collection

In the first stage, Kannada speech samples are collected using prerecorded audio files or microphone recordings. The speech signals are converted into a standard sampling format suitable for processing and recognition. The audio signals are typically resampled to 16 kHz mono format to ensure compatibility with the ASR model.

### 7.2 Stage 2: Far-Field Speech Simulation

To simulate practical far-field conditions, background noise is artificially added to the clean speech recordings. Additive White Gaussian Noise (AWGN) is used to create noisy speech signals with controlled Signal-to-Noise Ratio (SNR) values. This stage helps in analyzing the system performance under noisy environmental conditions.

### 7.3 Stage 3: Multi-Microphone Array Simulation

A virtual multi-microphone array setup is created to simulate speech capture from multiple microphones. In the proposed prototype, a 4-element microphone array is considered. Each microphone channel receives the speech signal along with different noise components, enabling spatial signal processing.

### 7.4 Stage 4: Beamforming-Based Speech Enhancement

Delay-and-Sum (DS) beamforming is applied to the microphone signals to enhance the desired speech signal. In this method, the captured microphone signals are aligned and averaged to suppress uncorrelated noise and improve speech clarity. The beamforming process enhances the Signal-to-Noise Ratio (SNR) of the speech signal before recognition.

The beamformed output can be mathematically represented as:

$$y(t) = \frac{1}{M} \sum_{m=1}^M x_m(t - \tau_m)$$

where:

- $y(t)$  represents the beamformed output signal



- $M$  represents the number of microphones
- $x_m(t)$  represents the signal received at the  $m^{th}$  microphone
- $\tau_m$  represents the time delay applied for signal alignment

## 7.5 Stage 5: Signal Preprocessing

After beamforming, the enhanced speech signal is processed using bandpass filtering and normalization techniques. The bandpass filter removes unwanted low-frequency and high-frequency noise components while preserving the important speech frequency range. Normalization is applied to maintain consistent signal amplitude levels.

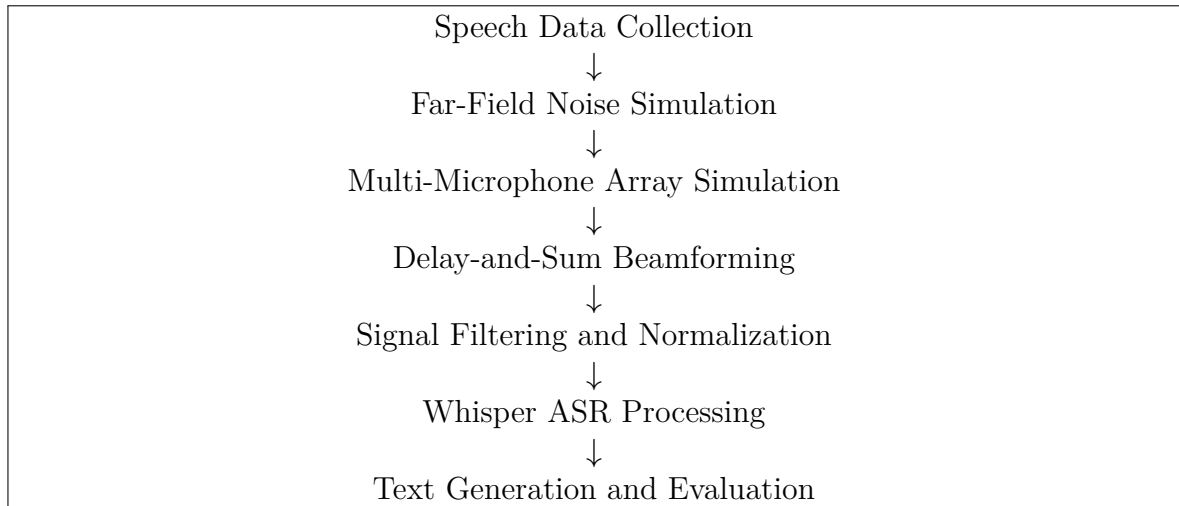
## 7.6 Stage 6: Speech Recognition

The processed speech signal is provided as input to the OpenAI Whisper Automatic Speech Recognition model. The ASR system converts the enhanced Kannada speech into text. The same recognition process is also performed on the noisy speech signal for comparison purposes.

## 7.7 Stage 7: Performance Evaluation

The final stage involves comparing the recognition performance of noisy speech and beamformed speech signals. Observations are made based on speech clarity, noise reduction, recognition stability, and transcription quality. Comparative analysis helps evaluate the effectiveness of beamforming for improving far-field speech recognition performance.

## 7.8 Overall Methodology Flow



## 8. System Architecture

---

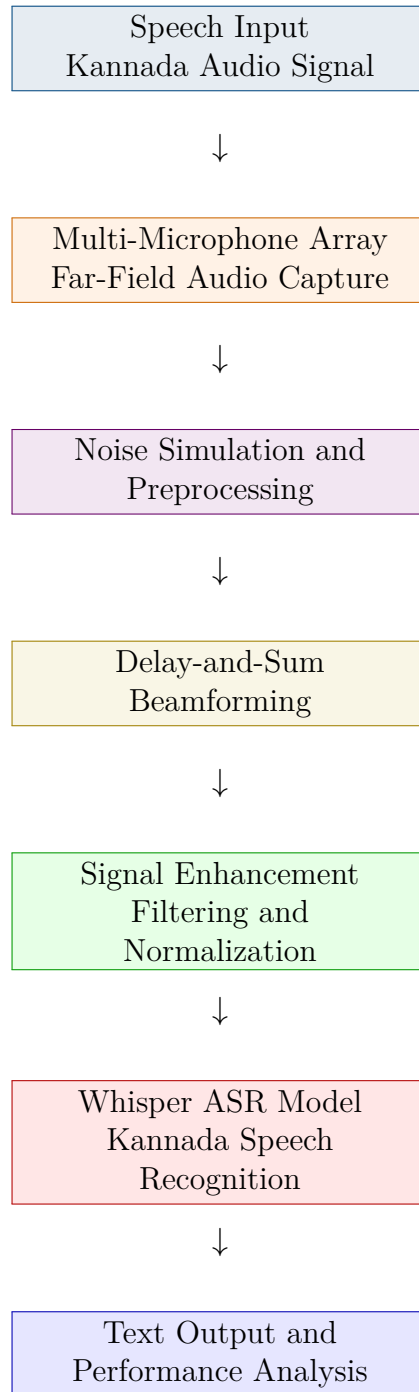
The proposed system architecture is designed to improve far-field speech recognition performance using beamforming-based speech enhancement techniques. The architecture consists of multiple interconnected modules that perform speech acquisition, preprocessing, signal enhancement, speech recognition, and performance evaluation.

The system begins with speech input captured from a distant speaker. The captured speech signal is affected by environmental noise, reverberation, and interference due to far-field recording conditions. To improve the quality of the speech signal, the system uses a simulated multi-microphone array and beamforming techniques before providing the enhanced signal to the Automatic Speech Recognition (ASR) model.

### 8.1 Architecture Components

1. **Speech Input Module**  
This module captures Kannada speech signals using prerecorded audio samples or microphone recordings. The speech signal acts as the primary input to the system.
2. **Multi-Microphone Array Simulation**  
A virtual 4-microphone array setup is created to simulate far-field speech acquisition. Multiple microphone channels are generated for performing spatial signal processing.
3. **Noise Simulation and Preprocessing Module**  
Background noise and interference are added to simulate real-world far-field conditions. Initial preprocessing operations such as resampling and signal conditioning are performed in this stage.
4. **Beamforming Module**  
Delay-and-Sum beamforming is applied to the microphone signals to enhance the desired speech signal and suppress unwanted environmental noise.
5. **Signal Enhancement Module**  
The beamformed speech signal is processed using bandpass filtering and normalization techniques to improve speech clarity and spectral quality.
6. **Automatic Speech Recognition Module**  
The enhanced speech signal is provided as input to the OpenAI Whisper ASR model, which converts Kannada speech into text format.
7. **Performance Evaluation Module**  
The system compares recognition outputs before and after beamforming enhancement to analyze the effectiveness of the proposed approach.

## 8.2 System Architecture Flow



## 8.3 Working Principle

The architecture operates by first capturing distant speech signals and simulating real-world noisy conditions. Multiple microphone channels are processed using beamforming techniques to improve the signal-to-noise ratio of the speech signal. The enhanced speech is then filtered and normalized before being processed by the ASR model. Finally, the generated text outputs are analyzed to evaluate the impact of beamforming on speech recognition performance.

## 9. Algorithms Used

---

The proposed system uses speech enhancement and speech recognition algorithms to improve the performance of far-field speech recognition for Kannada language speech. The implemented algorithms focus on reducing background noise, improving speech clarity, and generating more reliable recognition results in noisy environments.

### 9.1 1. Delay-and-Sum Beamforming Algorithm

Delay-and-Sum (DS) beamforming is one of the simplest and most widely used beamforming techniques for microphone array processing. In this method, speech signals received from multiple microphones are time-aligned and averaged to enhance the desired speech signal while suppressing unwanted noise and interference.

The received signal at each microphone can be represented as:

$$x_m(t) = s(t) + n_m(t)$$

where:

- $x_m(t)$  represents the signal captured at the  $m^{th}$  microphone
- $s(t)$  represents the desired speech signal
- $n_m(t)$  represents the noise component

The beamformed output signal is obtained by averaging the aligned microphone signals:

$$y(t) = \frac{1}{M} \sum_{m=1}^M x_m(t - \tau_m)$$

where:

- $y(t)$  is the beamformed output signal
- $M$  is the number of microphones
- $\tau_m$  is the time delay applied to the  $m^{th}$  microphone signal

The main advantage of Delay-and-Sum beamforming is its simplicity and ability to improve the Signal-to-Noise Ratio (SNR) by suppressing spatially uncorrelated noise.

### 9.2 2. Bandpass Filtering Algorithm

After beamforming, the speech signal is processed using a Butterworth bandpass filter to remove unwanted frequency components outside the speech range. The filter helps reduce low-frequency rumble and high-frequency noise while preserving important speech information.

The transfer function of the Butterworth filter is represented as:

$$|H(\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}}$$

where:

- $H(\omega)$  represents the filter response
- $\omega_c$  represents the cutoff frequency
- $n$  represents the filter order

The proposed system uses a bandpass range suitable for speech enhancement to improve speech intelligibility before recognition.

### 9.3 3. Signal Normalization

Signal normalization is applied after filtering to maintain a consistent amplitude range for all audio samples. This process improves stability during speech recognition and prevents amplitude variations from affecting the ASR model.

Normalization is performed using:

$$x_{norm}(t) = \frac{x(t)}{\max |x(t)|}$$

where:

- $x(t)$  is the input speech signal
- $x_{norm}(t)$  is the normalized signal

### 9.4 4. Whisper Automatic Speech Recognition Algorithm

The proposed system uses the OpenAI Whisper model for Automatic Speech Recognition (ASR). Whisper is a Transformer-based deep learning model trained on multilingual speech datasets. The model converts speech signals into text while supporting multiple languages, including Kannada.

The enhanced speech signal is provided as input to the Whisper ASR model, which generates the corresponding text output. The recognition results obtained from enhanced speech are compared with noisy speech outputs to evaluate the effectiveness of beamforming-based preprocessing.

### 9.5 5. Performance Evaluation Technique

The final stage involves comparing speech recognition outputs before and after beamforming enhancement. The evaluation focuses on:

- Speech clarity improvement
- Noise reduction effectiveness

- Recognition stability
- Quality of generated Kannada text
- Comparative analysis of noisy and enhanced speech signals

The evaluation helps determine whether beamforming techniques improve the overall performance of far-field speech recognition systems.

## 10. Dataset Description

---

The proposed project uses Kannada speech recordings for studying the effectiveness of beamforming techniques in far-field speech recognition systems. The dataset mainly consists of regional-language speech samples collected from prerecorded audio recordings and publicly available speech corpora.

### 10.1 Primary Dataset

The primary dataset used in the prototype consists of self-recorded Kannada speech samples collected for experimental analysis. The recordings include spoken sentences, general conversational speech, and short phrases captured using standard microphone devices.

The collected speech recordings are converted into a standard audio format suitable for signal processing and Automatic Speech Recognition (ASR). The audio files are typically processed using the following configuration:

- Sampling Frequency : 16 kHz
- Audio Format : WAV / MP3
- Channel Configuration : Mono Audio
- Language : Kannada
- Speech Type : Far-field and Noisy Speech

### 10.2 Far-Field Speech Simulation

Since real-world far-field recordings are difficult to collect in large quantities, the prototype simulates far-field conditions by artificially adding environmental noise to clean speech recordings. Additive White Gaussian Noise (AWGN) is used to create noisy speech signals with controlled Signal-to-Noise Ratio (SNR) levels.

The noisy speech generated during simulation helps evaluate the effectiveness of beamforming techniques under practical environmental conditions such as:

- Background noise
- Reverberation effects
- Interference from surrounding sounds
- Distant speech capture conditions

### 10.3 Multi-Microphone Data Generation

To simulate microphone array processing, multiple audio channels are generated from the original speech recordings. Each microphone channel contains slightly different noise characteristics, enabling the implementation of beamforming algorithms such as Delay-and-Sum beamforming.

The prototype uses a simulated 4-element microphone array configuration for spatial signal processing and speech enhancement experiments.

## 10.4 Secondary / Reference Datasets

In addition to self-recorded speech samples, publicly available Indian language speech datasets are considered for future evaluation and comparative analysis. These datasets include:

- AI4Bharat IndicVoices Dataset
- Mozilla Common Voice – Kannada Subset
- Regional-language speech samples from open-source repositories

These datasets provide additional speech diversity and support the evaluation of the proposed system across different speakers and recording conditions.

## 10.5 Purpose of the Dataset

The dataset is used for the following purposes:

1. Speech signal preprocessing and enhancement
2. Noise simulation and beamforming experiments
3. Testing the effectiveness of microphone array processing
4. Evaluating speech recognition performance before and after enhancement
5. Analyzing the feasibility of far-field Kannada speech recognition

The collected and simulated speech dataset helps validate the proposed prototype and provides a practical framework for analyzing beamforming-based far-field speech recognition systems for regional-language applications.



## 11. Experimental Setup

---

The experimental setup for the proposed project is designed to evaluate the performance of beamforming techniques for far-field speech recognition in Kannada language applications. The setup includes speech signal acquisition, noise simulation, beamforming-based enhancement, and Automatic Speech Recognition (ASR) evaluation using software-based implementation tools.

### 11.1 System Configuration

The experiments are carried out using a computer-based software environment with simulated microphone array processing. The complete implementation is developed using Python programming and speech processing libraries.

### 11.2 Hardware Requirements

The following hardware components are used for conducting the experiments:

- Laptop / Desktop Computer
- Standard Audio Input Microphone
- Simulated 4-Element Microphone Array
- Internet-enabled environment for downloading ASR models
- GPU-enabled execution environment (Google Colab T4 GPU optional)

### 11.3 Software Requirements

The following software tools and libraries are used in the implementation:

- Python 3.10+
- Google Colab / Jupyter Notebook
- OpenAI Whisper ASR Model
- NumPy for numerical computations
- SciPy for signal filtering and processing
- Librosa for audio analysis
- Matplotlib for waveform and spectrogram visualization
- SoundFile and FFmpeg for audio handling

### 11.4 Audio Processing Configuration

The speech signals are processed using the following audio configuration:

| Parameter                | Configuration                 |
|--------------------------|-------------------------------|
| Sampling Frequency       | 16 kHz                        |
| Audio Format             | WAV / MP3                     |
| Audio Channel            | Mono                          |
| Microphone Configuration | Simulated 4-Microphone Array  |
| Beamforming Technique    | Delay-and-Sum Beamforming     |
| Noise Type               | Additive White Gaussian Noise |
| Signal-to-Noise Ratio    | 5 dB                          |
| Filtering Technique      | Butterworth Bandpass Filter   |
| Speech Recognition Model | OpenAI Whisper Medium         |
| Language Processed       | Kannada                       |

## 11.5 Experimental Procedure

The experiments are conducted through the following sequence of steps:

- Step 1: Kannada speech samples are collected and converted into the required audio format.
- Step 2: Background noise is added to simulate far-field speech conditions.
- Step 3: Multiple microphone channels are generated for microphone array simulation.
- Step 4: Delay-and-Sum beamforming is applied to enhance the speech signal.
- Step 5: The enhanced signal is filtered and normalized for ASR processing.
- Step 6: Both noisy and enhanced speech signals are transcribed using the Whisper ASR model.
- Step 7: The generated text outputs are compared to evaluate speech enhancement performance.

## 11.6 Performance Evaluation Parameters

The effectiveness of the proposed system is evaluated using the following observations:

- Speech clarity improvement
- Reduction in background noise
- Stability of speech recognition output
- Quality of generated Kannada transcription

- Comparison between noisy and enhanced speech signals

## 11.7 Expected Experimental Outcome

The experimental setup is expected to demonstrate that beamforming-based speech enhancement improves the quality of far-field Kannada speech signals and provides more reliable speech recognition results compared to noisy single-channel speech. The setup also validates the feasibility of implementing practical far-field ASR systems for regional-language applications using software-defined microphone array processing techniques.

## 12. Results and Analysis

---

The proposed prototype was tested to analyze the effectiveness of beamforming techniques for improving far-field speech recognition performance in Kannada language applications. The experiments focused on comparing noisy speech signals with beamformed and enhanced speech signals using qualitative observations and speech recognition outputs.

### 12.1 Speech Enhancement Analysis

The experimental observations showed that the application of Delay-and-Sum beamforming improved the overall quality of the speech signal captured under far-field noisy conditions. Background noise and unwanted interference were reduced after beamforming and filtering operations.

The enhanced speech signal demonstrated:

- Improved speech clarity
- Reduced environmental noise
- Better signal stability
- Improved intelligibility of Kannada speech
- More consistent waveform characteristics

Visual analysis using waveform and spectrogram representations also indicated clear improvement in speech quality after beamforming-based enhancement.

### 12.2 Speech Recognition Analysis

The processed speech signals were evaluated using the OpenAI Whisper ASR model. Recognition outputs obtained from noisy speech signals were compared with the outputs generated from enhanced beamformed speech signals.

The noisy speech signals produced partially distorted and unstable recognition results because of the presence of noise and interference. In contrast, the beamformed speech signals generated more coherent and stable Kannada text outputs.

## 12.3 Comparative Analysis

| Parameter                       | Before Enhancement  | After Enhancement  |
|---------------------------------|---------------------|--------------------|
| Speech Signal Quality           | Noisy and Distorted | Improved and Clear |
| Background Noise                | High                | Reduced            |
| Speech Intelligibility          | Moderate            | Improved           |
| Beamforming Applied             | No                  | Yes                |
| Signal Filtering                | Not Applied         | Applied            |
| Recognition Stability           | Unstable            | More Stable        |
| Kannada Text Quality            | Partially Distorted | More Coherent      |
| Far-Field Performance           | Affected by Noise   | Improved           |
| Overall Recognition Reliability | Moderate            | Better             |

## 12.4 Observations

1. Delay-and-Sum beamforming successfully reduced the effect of environmental noise present in the speech signal.
2. The enhanced speech signal showed better clarity compared to the noisy single-channel speech signal.
3. Speech recognition outputs generated after enhancement were more stable and understandable.
4. The combination of beamforming, filtering, and normalization improved the performance of the ASR system under far-field conditions.
5. The prototype demonstrated the feasibility of using beamforming techniques for regional-language far-field speech recognition applications.

## 12.5 Limitations Observed

Although the prototype demonstrated encouraging results, some limitations were observed during experimentation:

- A simulated microphone array was used instead of a physical hardware array.
- The system performance still depends on the quality of the input recording and noise conditions.
- Only Delay-and-Sum beamforming was implemented in the current prototype.
- Real-time speech processing was not included in the present phase of the project.

## 12.6 Conclusion from Experimental Results

The experimental results indicate that beamforming-based preprocessing can improve the quality of far-field Kannada speech signals and provide better recognition performance compared to noisy speech signals. The prototype validates the feasibility of integrating microphone array processing and speech recognition techniques for regional-language speech applications in practical environments.

## 13. Future Scope

---

The proposed project demonstrates the feasibility of improving far-field speech recognition using beamforming techniques for Kannada language applications. Although the current prototype provides encouraging results, several enhancements and extensions can be implemented in future phases of the work.

1. Implementation of Advanced Beamforming Techniques

The present prototype mainly uses Delay-and-Sum beamforming for speech enhancement. In future work, advanced techniques such as Minimum Variance Distortionless Response (MVDR), adaptive beamforming, and neural beamforming methods can be implemented to achieve better noise suppression and speech clarity.

2. Real Hardware Microphone Array Integration

The current system uses a software-simulated microphone array. Future development can include the use of real multi-microphone hardware arrays for practical far-field speech acquisition and real-time experimentation.

3. Real-Time Speech Recognition System

The prototype currently processes prerecorded audio samples. Future improvements can focus on developing a real-time far-field speech recognition system capable of processing live speech input continuously.

4. Support for Multiple Regional Languages

The system can be extended to support additional Indian regional languages such as Telugu, Tamil, Hindi, and Malayalam, thereby improving multilingual accessibility for voice-enabled technologies.

5. Deep Learning-Based Speech Enhancement

Future work can integrate deep neural network-based speech enhancement models and Transformer-based beamforming approaches to improve robustness in highly noisy and reverberant environments.

6. Improved Noise and Reverberation Handling

Additional preprocessing techniques such as dereverberation, echo cancellation, and adaptive noise suppression can be incorporated to improve system performance under complex acoustic conditions.

7. Performance Evaluation Using Larger Datasets

Future experiments can utilize large-scale public speech datasets and real-world recordings to evaluate the scalability and generalization capability of the system.

8. Integration with Smart Applications

The developed system can be integrated into practical applications such as virtual assistants, smart classrooms, voice-controlled devices, assistive communication

systems, and public information systems.

#### 9. Optimization for Embedded and Edge Devices

The proposed framework can be optimized for deployment on embedded systems, mobile devices, and edge-computing platforms to enable low-cost and portable speech recognition solutions.

The future enhancements can help develop a more robust, scalable, and practical far-field speech recognition system capable of supporting real-world regional-language applications with improved accuracy and reliability.



## 14. Conclusion

---

The proposed project successfully demonstrates the feasibility of improving far-field speech recognition using beamforming techniques for Kannada language applications. The work focused on enhancing distant speech signals affected by background noise and environmental interference before processing them through an Automatic Speech Recognition (ASR) system.

The implemented prototype combined microphone array simulation, Delay-and-Sum beamforming, signal preprocessing, filtering, and Whisper-based speech recognition within a unified framework. Experimental analysis showed that beamforming-based speech enhancement improved speech clarity, reduced background noise, and produced more stable recognition outputs compared to noisy speech signals.

The study also highlighted the importance of preprocessing techniques in improving the performance of far-field speech recognition systems operating under practical environmental conditions. The results indicate that microphone array processing and beamforming can significantly contribute towards enhancing regional-language speech recognition performance.

In addition, the project emphasizes the growing need for robust speech technologies for Indian regional languages such as Kannada. Most existing speech recognition research focuses mainly on high-resource languages, while regional-language far-field ASR systems still require further development and optimization.

Overall, the prototype validates the effectiveness of beamforming-based speech enhancement for far-field speech recognition and provides a strong foundation for future work involving advanced beamforming methods, real-time processing, and practical deployment in voice-enabled applications.

## 15. References

---

- [1] J. Chen, J. Benesty, and Y. Huang, “Microphone Array Signal Processing for Speech Enhancement,” *IEEE Signal Processing Magazine*, vol. 23, no. 5, pp. 26–34, Sept. 2006.
- [2] S. Doclo and M. Moonen, “Design of Broadband Beamformers Robust Against Gain and Phase Errors in Microphone Array Characteristics,” *IEEE Transactions on Signal Processing*, vol. 51, no. 10, pp. 2511–2526, Oct. 2003.
- [3] X. Xiao et al., “Deep Beamforming Networks for Multi-Channel Speech Recognition,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2016, pp. 5745–5749.
- [4] Z. Meng, S. Watanabe, J. R. Hershey, and H. Erdogan, “Deep Long Short-Term Memory Adaptive Beamforming Networks for Multichannel Robust Speech Recognition,” in *Proc. IEEE ICASSP*, 2017.
- [5] W. He, L. Lu, B. Zhang, J. Mahadeokar, K. Kalgaonkar, and C. Fuegen, “Spatial Attention for Far-Field Speech Recognition with Deep Beamforming Neural Networks,” in *Proc. IEEE ICASSP*, 2020.
- [6] W. Zhang, A. S. Subramanian, X. Chang, S. Watanabe, and Y. Qian, “End-to-End Far-Field Speech Recognition with Unified Dereverberation and Beamforming,” in *Proc. Interspeech*, 2020.
- [7] K. Kumatani, M. Wu, S. Sundaram, N. Strom, and B. Hoffmeister, “Multi-Geometry Spatial Acoustic Modeling for Distant Speech Recognition,” in *Proc. IEEE ICASSP*, 2019.
- [8] D. Wang, Z. Chen, and T. Yoshioka, “Neural Speech Separation Using Spatially Distributed Microphones,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 715–726, 2020.
- [9] T. Yoshioka and M. Delcroix, “Neural Network Based Beamforming for Far-Field Speech Recognition,” *IEEE Signal Processing Letters*, vol. 24, no. 6, pp. 855–859, 2017.
- [10] C. Boeddeker, H. Erdogan, T. Yoshioka, and R. Haeb-Umbach, “Exploring Practical Aspects of Neural Mask-Based Beamforming for Far-Field Speech Recognition,” in *Proc. IEEE ICASSP*, 2018.
- [11] S. Araki et al., “Online Meeting Recognition in Noisy Environments with Time-Frequency Mask Based MVDR Beamforming,” in *Proc. HSCMA*, 2017.
- [12] H. Chen, P. Zhang, and Y. Yan, “Multi-Talker MVDR Beamforming Based on Extended Complex Gaussian Mixture Model,” *IEEE Signal Processing Letters*, vol. 27, pp. 1829–1833, 2020.

- [13] Z. Ni, F. Grezes, V. A. Trinh, and M. I. Mandel, “Improved MVDR Beamforming Using LSTM Speech Models to Clean Spatial Clustering Masks,” in Proc. IEEE ICASSP, 2021.
- [14] B. J. Cho and H. M. Park, “Convolutional Maximum-Likelihood Distortionless Response Beamforming With Steering Vector Estimation for Robust Speech Recognition,” IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 29, pp. 2489–2500, 2021.
- [15] L. Mosner, P. Matejka, O. Novotny, and J. Cernocky, “Dereverberation and Beamforming in Far-Field Speaker Recognition,” in Proc. IEEE ICASSP, 2018.
- [16] C. Li and T. Wang, “The ZTSpeech System for CHiME-5 Challenge: A Far-Field Speech Recognition System with Front-End and Robust Back-End,” in Proc. CHiME Workshop, 2018.
- [17] T. Ochiai et al., “Beam-TasNet: Time-Domain Audio Separation Network Meets Frequency-Domain Beamformer,” in Proc. IEEE ICASSP, 2020.
- [18] M. I. Mandel and J. P. Barker, “Multichannel Spatial Clustering for Robust Far-Field Automatic Speech Recognition in Mismatched Conditions,” in Proc. Interspeech, 2016.
- [19] D. Yu and L. Deng, Automatic Speech Recognition: A Deep Learning Approach. London, U.K.: Springer, 2015.
- [20] D. Jurafsky and J. H. Martin, Speech and Language Processing, 3rd ed. Pearson, 2023.
- [21] L. R. Rabiner and B. H. Juang, Fundamentals of Speech Recognition. Englewood Cliffs, NJ, USA: Prentice Hall, 1993.
- [22] M. Brandstein and D. Ward, Microphone Arrays: Signal Processing Techniques and Applications. Berlin, Germany: Springer, 2001.
- [23] A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision,” arXiv preprint arXiv:2212.04356, 2022.
- [24] AI4Bharat, “IndicVoices: A Large-Scale Multilingual Indian Speech Dataset,” 2025. [Online]. Available: <https://ai4bharat.org/>. Accessed: May 2026.
- [25] Mozilla Foundation, “Common Voice Dataset,” 2025. [Online]. Available: <https://commonvoice.mozilla.org/>. Accessed: May 2026.

---

End of Synopsis Report